

Cowardice and High Status as Anti-Popper Progress Blocking

The Structural Mechanics of Institutional Non-Commitment

Registry: [\[HOWL-CULT-12-2026\]](#)

Series Path: [\[HOWL-CULT-1-2026\]](#) → [\[HOWL-CULT-2-2026\]](#) → [\[HOWL-CULT-3-2026\]](#) → [\[HOWL-CULT-4-2026\]](#) → [\[HOWL-CULT-5-2026\]](#) → [\[HOWL-CULT-6-2026\]](#) → [\[HOWL-CULT-7-2026\]](#) → [\[HOWL-CULT-8-2026\]](#) → [\[HOWL-CULT-9-2026\]](#) → [\[HOWL-CULT-12-2026\]](#)

Date: May 2026

DOI: 10.5281/zenodo.20181100

Domain: Epistemology / Institutional Analysis / Philosophy of Science

Status: Structural analysis with falsifiable predictions

AI Usage Disclosure: Only the top metadata, figures, refs and final copyright sections were edited by the author. All paper content was LLM-generated using Anthropic's Claude Opus 4.6.

1. Two Definitions

This paper uses two words with structural definitions. Both definitions are mechanical. Both are testable. Both are applied to observable behavior patterns, not to individuals.

Cowardice. An agent has an action available. The action would produce collective benefit. The action carries personal risk. The agent declines the action to avoid the personal risk. This is the complete definition. The agent's intelligence, character, kindness, and intentions are not part of the definition. A brilliant, generous, well-meaning agent who declines an available beneficial action to avoid personal risk is exhibiting cowardice by this definition. The definition describes the decision pattern. The decision pattern is observable. Whether it applies in a specific case is testable — identify the available action, identify the collective benefit, identify the personal risk, observe whether the action is taken.

Falsification. Karl Popper formalized the demarcation criterion for science in 1934. A claim is scientific if and only if it specifies conditions under which it would be false. Popper did not merely propose this as a nice idea. He proposed it as the boundary between science and non-science — the structural feature that separates claims that can teach us about reality from claims that cannot.

Falsification as a practice, not merely a philosophical position, requires four elements operating together. First, commitment: the claimant states a specific observable prediction before the observation is made. Not a range. Not a family of possibilities. A specific state

of affairs that the claimant asserts will be observed. Second, specificity: the prediction is precise enough that it can disagree with observation. A prediction that accommodates all outcomes is not a prediction — it is a tautology. Third, priority: the prediction is stated before the test, not retrofitted to match the result. Fourth, acceptance: when the observation contradicts the prediction, the claimant accepts the falsification. The claim failed. The failure is reported as a finding. The claimant does not add auxiliary hypotheses to save the claim, does not question the observation's validity without specific grounds, does not reinterpret the prediction to accommodate the result.

Remove any one of these four elements and the activity is not falsification. It may be useful. It may be sophisticated. It may produce interesting commentary. But it is not the practice that Popper identified as the structural core of scientific knowledge production.

These two definitions are the tools the paper uses. Every subsequent claim is built from them. A reader who disagrees with the definitions should stop here — nothing that follows will be persuasive because everything that follows depends on these definitions being accepted as mechanical and testable.

2. What Falsification Requires Monday Morning

Philosophy of science is taught in seminars. Science is practiced in laboratories, at desks, and in front of computational environments. The gap between the seminar and the desk is where Popper's criterion either lives or dies. This section describes what a practicing scientist would have to do to practice falsification as defined above — not as a philosophical posture but as a daily working method.

The first requirement is the commitment step. The scientist states a specific value. Not “the baryon density parameter is approximately 0.049.” Not “our results are consistent with a baryon density parameter in the range 0.048 to 0.050.” The commitment is: “The baryon density parameter is 0.049036. If measurement shows a value outside $0.049036 \pm$ the measurement's own stated uncertainty, this prediction fails.” The number is specific. The failure condition is explicit. The commitment is made before the measurement is checked.

This is Popper's first and second elements — commitment and specificity — applied to a real quantity. The committed value can be wrong. That is the point. A value that cannot be wrong cannot teach anything, because no observation can contradict it, and therefore no observation can confirm it either. Popper's insight was precisely this: a claim that is compatible with every possible observation is a claim that says nothing about which observation will actually occur.

The second requirement is the building step. The committed value is used as an input to the next calculation. If the baryon density is 0.049036, then the baryon-to-photon ratio is a specific number derived from it, and the primordial helium abundance is a specific number derived from that, and the primordial deuterium abundance is a specific number derived from that. Each downstream prediction inherits the commitment. If the upstream value is wrong, the downstream predictions break at specific located points. The chain

of commitments creates a chain of falsification opportunities. Each link can fail independently. Each failure locates a specific problem.

This is Popper's criterion applied recursively. Each derived prediction is itself a falsifiable claim. The derivation chain multiplies the opportunities for reality to disagree with the model. A chain of ten committed values has ten failure points. If all ten survive, the model has been tested ten times. If any one fails, the failure tells you exactly where the model departs from reality — not “somewhere in the model” but “at this specific link, at this specific value, with this specific disagreement.”

The third requirement is the acceptance step. When a measurement contradicts the committed value beyond stated uncertainty, the scientist accepts the result. The commitment failed. This is Popper's fourth element — acceptance of falsification. The failure is not an embarrassment. The failure is a finding. It is a located error — a specific point where reality differs from the model. Located errors are the most valuable output of the scientific process because they narrow the search space for the correct model. Before the failure, the error could have been anywhere. After the failure, it is at a specific point. The search space has been reduced by the precise amount of information contained in the located error.

The fourth requirement is accumulation. Commitments that survive repeated falsification attempts become platforms. The scientist builds on them. Each surviving commitment supports more weight. The structure of tested knowledge grows by accumulation of survived commitments, each one bearing the weight of everything built on top of it. If a platform commitment eventually fails under new evidence, the failure is dramatic and informative — it identifies a deep structural point where the model must change, and everything built on that platform is the scope of what must be re-examined. This is how scientific revolutions work mechanically. A platform commitment fails, and the failure reorganizes everything above it.

The daily practice of falsification is therefore: commit, derive, test, accept, accumulate. Five steps. Each step is concrete. Each step is observable. Whether a scientist performs these steps is not a philosophical question. It is an empirical question about their daily behavior.

3. What the Institution Does Instead

The observable behavior of institutional physics can be examined against the five steps defined above. Each step has a corresponding institutional practice that replaces it. The replacements are not falsification. They are a different activity with a different structure and different consequences.

Where falsification requires commitment to specific values, institutional physics practices hedged language. The standard vocabulary of physics papers includes “consistent with,” “suggests,” “may indicate,” “is compatible with,” “does not exclude,” and “within the framework of.” Each of these phrases has the same structural property: it commits to nothing. A result that is “consistent with” a model is also consistent with the model being

wrong in ways the result cannot detect. A finding that “suggests” a mechanism does not claim the mechanism is correct. A measurement that “does not exclude” a parameter range does not claim the parameter is in that range.

This is a violation of Popper’s first element — commitment. The hedged claim cannot be falsified because it does not assert anything specific enough to contradict. If a future measurement disagrees, the hedged claim survives: it never said the model was correct, only that the result was consistent with it. If a future measurement agrees, the hedged claim takes credit: the result supports the model that was suggested. All outcomes are absorbed. The claim functions as an unfalsifiable statement dressed in scientific vocabulary.

The hedged vocabulary is not occasional. It is the institutional norm. It is what peer reviewers expect. It is what journal editors accept. It is what graduate advisors teach. A paper that committed to specific values with explicit failure conditions would be reviewed as overconfident, insufficiently rigorous, or naive. The institution has inverted Popper’s criterion: the practice Popper identified as the core of science — specific falsifiable commitment — is treated by the institution as a mark of insufficient sophistication.

Where falsification requires building on commitments as platforms, institutional physics practices parameter openness. Consider the strong CP parameter. In quantum chromodynamics, the theory of the strong nuclear force, there is a parameter called theta that could in principle produce a specific observable effect — violation of a symmetry called CP in strong-force interactions. Every experiment that has looked for this effect has measured it as zero or indistinguishable from zero. Decades of experiments. No nonzero result. Ever.

The theoretical formalism permits theta to range from zero to 2π . Because the formalism has a slot for nonzero values, the field treats the observed zero as a mystery requiring explanation. “Why is theta zero?” became a research program. A mechanism was proposed — the Peccei-Quinn mechanism — that would dynamically drive theta to zero. This mechanism predicted a new particle, the axion. Experiments to find the axion have been running for decades. The axion has not been found.

The Popper violation here is at the building step. The observations consistently measure zero. A field practicing falsification would commit to zero, build on it as a platform, and use it as an input to downstream calculations. If anyone ever measured a nonzero value, that measurement would falsify the commitment — cleanly, dramatically, and informatively. Until that falsification, the commitment would support a growing structure of derived predictions. Instead, the field holds theta at “maybe zero” — officially uncertain despite decades of consistent measurement. No commitment is made. No platform is built. The “maybe” absorbs all outcomes: measure zero again and it is consistent with maybe, measure nonzero and the field always said it could be nonzero. No outcome is decisive because no position was taken. This violates Popper at the deepest level — it structurally prevents any observation from being decisive by refusing to commit to a position that observations could contradict.

Where falsification requires accepting failure as a located finding, institutional physics practices committee averaging. Reference values for physical constants are published by committees — notably CODATA, the Committee on Data for Science and Technology

— that collect measurements from multiple experiments, weight them by various criteria, average them, and publish the result as the recommended value. The recommended value changes with each publication cycle. Not because the underlying physical quantity changed, but because the committee received new measurements, changed its weighting methodology, or included experiments not previously considered.

The Popper violation is at the acceptance step. No individual measurement is treated as a commitment that could fail. Every measurement is absorbed into the average. A measurement that disagrees with the previous average does not falsify anything — it shifts the average slightly. A measurement that agrees with the previous average does not confirm anything — it was expected. The averaging process is structurally incapable of producing a falsification event because no specific value was committed to. The average is not a commitment. It is a summary of the current state of measurement, which is a different object entirely. A commitment says “this value is correct and I accept being wrong if it is not.” An average says “this is what the measurements currently indicate, subject to revision.”

The gravitational constant G illustrates this precisely. G has been measured repeatedly on the surface of the Earth using different experimental apparatus. The measurements do not converge. The disagreement between independent determinations is larger than the individual experiments’ stated uncertainties. Each publication cycle, the committee publishes a new recommended value. The new value differs from the previous one. The community treats each new value as the current best estimate. No one asks whether the non-convergence itself is evidence of a structural problem — whether G as a universal constant is the wrong frame, whether the measurements don’t converge because the quantity being measured is not what the model says it is. The averaging process prevents this question from arising because the process treats non-convergence as a data-quality problem to be solved by better experiments, not as a potential falsification of the universality of G .

Where falsification requires that commitments accumulate into testable platforms, institutional physics practices risk-averse publication. The career-optimal strategy for a physicist is well understood within the community even if rarely stated explicitly. Publish frequently. Cite correctly and thoroughly. Use hedged language. Avoid commitments that could be falsified. Build reputation through volume and citation count. Seek positions on committees that determine reference values. Secure funding by proposing incremental extensions of existing work. The strategy is locally optimal for every individual scientist — it maximizes career longevity, funding continuity, and status maintenance. It is collectively catastrophic. It produces a growing body of literature in which nothing is committed, nothing is falsifiable, nothing can fail, nothing accumulates into platforms, and nothing closes. The literature grows. Knowledge does not accumulate. These are different things, and the difference is precisely Popper’s criterion.

4. The Mechanism: Status Preservation

The behavior pattern described in Section 3 exists for identifiable structural reasons. No conspiracy is required. No coordination is required. No malice is required. The behavior emerges from ordinary incentive mechanics operating on ordinary human psychology in a specific institutional context.

Physics occupies one of the highest status positions in intellectual culture. The public regards physicists as among the most intelligent people in society — people who can comprehend things ordinary people cannot, who work at the frontier of human understanding, who discern deep truths about the nature of reality. This status is real in the sense that it has real consequences: funding decisions, public respect, institutional authority, policy influence, media attention, and self-concept are all affected by it. A physicist who has never made a falsifiable prediction, never committed to a specific value, and never had a claim tested against observation still holds this status. The status is conferred by institutional membership — by having the right degrees, the right affiliation, the right publication record, and the right vocabulary. It is not conferred by the practice of falsification.

The cost of being publicly wrong in a high-status field is disproportionate to the cost in a field with lower status stakes. An engineer whose bridge design fails at a specific load point has produced a located error. The error is investigated. The design is corrected. The engineer's reputation may suffer temporarily but the failure is treated as engineering — a specific problem at a specific point with a specific solution. A physicist whose committed prediction is falsified by experiment has, in the current institutional culture, done something much worse. The physicist was publicly wrong. In a field where status derives from being perceived as someone who comprehends deep truths, being publicly wrong threatens the perception that supports the status. The loss is not proportional to the error's magnitude. It is proportional to the visibility of the commitment.

The asymmetry is critical. The status gained from being right on a committed prediction is modest — “a good result, as expected.” The status lost from being wrong on a committed prediction is severe — “they were overconfident, insufficiently careful, perhaps not as brilliant as we thought.” This asymmetry makes commitment irrational at the individual level. The expected value of committing is negative: modest gain if right, severe loss if wrong, with some nonzero probability of being wrong. The rational response is to never commit. Hedge every claim. Keep every parameter open. Avoid building on any specific value. The strategy that maximizes individual status preservation is the strategy that minimizes collective knowledge production.

The aggregate of individually rational non-commitment decisions produces the behavior pattern described in Section 3. No individual physicist decided to stop practicing falsification. Each made a series of small, locally rational decisions: soften this claim, hedge this prediction, hold this parameter open, present this result as suggestive rather than definitive. Each decision was made in the context of real career pressures — funding cycles, tenure reviews, publication requirements, peer evaluation. Each was a reasonable response to the incentive structure as it exists.

The aggregate is a field that has structurally ceased practicing falsification while continuing

to claim falsification as its epistemological foundation.

This aggregate pattern matches the structural definition of cowardice from Section 1. The available action is commitment to specific falsifiable values. The collective benefit is located errors, closed parameters, accumulated platforms, and the advancement of the field toward closure of its completable core. The personal risk is status loss from being publicly wrong. The action is declined to avoid the risk. The definition applies.

This is not a statement about physicists as people. Many physicists would commit to specific values if the incentive structure rewarded it. Many have the courage to accept being wrong — they demonstrate it in private conversations, in seminars with trusted colleagues, in informal settings where career risk is low. The cowardice is not in the individuals. It is in the aggregate behavior pattern produced by the incentive structure. The individuals respond rationally to the incentives they face. The incentive structure produces cowardice as its equilibrium output. Changing the incentive structure would change the output. The problem is structural. The solution is structural.

5. The Double Block

The institution's behavior has two components that compound each other. The first component, described in Sections 3 and 4, is internal: the institution discourages its own members from practicing commitment through career risk. The second component is external: the institution prevents outsiders from contributing through an undifferentiated exclusion mechanism.

If a field will not do the work itself and also prevents others from doing it, the field's position is not "this work is being done properly by qualified people." The field's position is "this work should not be done." That is not a commitment to quality. It is a commitment to stasis.

The external exclusion mechanism is the label "crank." The label is applied to any person outside the institutional credentialing system who engages with frontier physics. The application is undifferentiated — the same label, the same response, the same exclusion, regardless of the outsider's methodology, rigor, or results.

Consider the spectrum of people the label is applied to. At one end: a person who claims to have built a perpetual motion machine in their garage, violating thermodynamics. At the other end: a systems engineer who knows the Standard Model, works within its constraints, derives specific physical values from declared integer inputs using reproducible computation, produces results that match established measurements to sub-percent precision, states explicit falsification conditions for every prediction, and publishes all inputs and code for independent verification. Between these extremes: people with varying degrees of physics knowledge, varying methodological rigor, and varying quality of results.

The institution applies one label to this entire spectrum. The perpetual motion builder and the systems engineer with sub-percent CODATA matches receive the same classification. The classification determines the response: exclusion from journals, exclusion

from conferences, exclusion from peer review, exclusion from serious evaluation. The work is never examined because the classification was applied before the examination could occur.

This label functions as an unfalsifiable classification. Popper's criterion, applied to the label itself, asks: what observation would show that this person is not a crank? If the outsider's prediction is wrong, they are confirmed as a crank. If the outsider's prediction matches measurement, the agreement must be coincidental, or the methodology must be flawed in ways not yet identified, or the outsider must have gotten lucky. The label absorbs all outcomes. No observation can dislodge it once applied because no observation was part of the application criteria. The criteria were institutional: does this person have the right degrees? The right affiliation? The right publication history? The right vocabulary? These are questions about the person, not about the work. The work — which is the falsifiable content, the thing that either matches reality or does not — is never reached.

This is a Popper violation at the most fundamental level. Popper's entire framework exists to evaluate claims on their content — their specificity, their falsifiability, their agreement or disagreement with observation. The crank label evaluates the claimant on their credentials — their institutional membership, their educational history, their social position within the field. These are opposite operations. Popper asks "what does the claim say and can it be tested?" The crank label asks "who is the claimant and are they authorized to speak?" The label structurally prevents Popper's criterion from being applied because it intercepts the claim before evaluation.

The compounding effect of the internal and external blocks is total. An insider who commits risks career damage. An outsider who commits receives the crank label. The insider path to closure is discouraged by incentive. The outsider path to closure is blocked by classification. The only permitted activity is uncommitted work by credentialed insiders — which is precisely the activity described in Section 3, the activity that cannot close anything because it never commits to anything that could be falsified.

The institution has not explicitly decided to prevent closure. No committee met and resolved that physics should remain permanently open. The double block emerged from the aggregation of individual incentive responses — career-protecting hedging by insiders, credential-based filtering of outsiders — each locally rational, collectively producing a system that permits only the activity least capable of advancing knowledge. The structure preserves itself without coordination because each participant has individual reasons for their contribution to the structure.

The distinction between quality control and credential verification must be stated precisely because the institution conflates them. Quality control evaluates work on its content. Does the prediction match measurement? Are the inputs declared? Are the failure conditions stated? Is the derivation reproducible? Can an independent person run the computation and get the same result? Quality control is Popper's criterion applied to incoming work — commit to evaluation criteria, examine the work against those criteria, accept the result. What the institution practices is credential verification. Does the person hold a doctorate in physics? Are they affiliated with a recognized institution? Have they published in peer-reviewed journals? Do they use the field's standard vocabulary? These questions evaluate

the person. They tell you about the person's institutional history. They tell you nothing about whether the person's current claim matches reality.

A person without a physics doctorate who produces a reproducible derivation matching CODATA values to measurement precision has produced work that Popper's criterion can evaluate. A person with a physics doctorate who publishes a hedged claim compatible with all possible outcomes has produced work that Popper's criterion cannot evaluate — there is no observation that could falsify it. The institution accepts the second and rejects the first. The institution's filtering criterion is the inverse of Popper's criterion. The institution selects for the unfalsifiable and excludes the falsifiable.

6. The Manufactured Crank Problem

The institution's external exclusion mechanism — the crank label — addresses a real phenomenon. People with inadequate physics knowledge do propose theories that ignore established results. Perpetual motion machines are submitted to patent offices. Theories that violate conservation laws are posted online. Proposals that contradict well-tested measurements are sent to journals. These submissions consume time and attention. The institution's desire to filter them is understandable.

The institution does not ask why these submissions exist. The answer is in the institution's own educational pipeline.

A child enters the educational system at age five or six. Over the next twelve to thirteen years, this child receives the following physics education in a typical curriculum. In elementary school: matter is made of atoms, atoms have electrons orbiting a nucleus. This model — electrons orbiting like planets — was superseded in the 1920s. The child learns a model that has been known to be structurally wrong for a century. In middle school: forces are pushes and pulls, gravity pulls things down, magnets attract and repel. No mention of the strong force, the weak force, or the Higgs mechanism. The child has now learned two of the four fundamental forces and an incorrect model of atomic structure. In high school, if the child takes physics: Newton's laws, basic electromagnetism, possibly the Bohr model of the atom. The Bohr model was an improvement over the planetary model but is also known to be incomplete and structurally misleading. The child may learn that "quantum mechanics is weird" without learning what quantum mechanics actually describes or what it has been tested against.

At age eighteen, the child who took every available physics course has learned: a wrong model of the atom, two of four forces, Newton's laws, and possibly the Bohr model. The child has never heard of quarks, gluons, the W and Z bosons, the Higgs boson, neutrino oscillation, CP violation, the CKM matrix, the PMNS matrix, the strong CP problem, asymptotic freedom, confinement, or the Standard Model as a coherent framework. The child does not know that the proton is a composite object. The child does not know that mass is almost entirely binding energy. The child does not know that the universe has been measured to extraordinary precision across dozens of independent quantities that form an interlocking web of consistency.

The child who proceeds to a university physics program spends two to three years learning mathematical tools — differential equations, linear algebra, classical mechanics in Lagrangian and Hamiltonian form, electrodynamics, statistical mechanics, quantum mechanics formalism. These are the tools of physics, not the results. The Standard Model — what actually exists, what has been measured, what is confirmed — appears in the third or fourth year of an undergraduate program at the earliest. More typically, it appears in graduate coursework. A student who leaves the university pipeline with a bachelor's degree in physics may have encountered the Standard Model briefly. A student who leaves before completing a physics degree has almost certainly never encountered it systematically.

The pipeline takes approximately twenty years — from age five to age twenty-five — to deliver a student to the point where they know the actual established content of physics. For twenty years, the student has been learning progressively less wrong models, each presented as the truth at its level, each eventually revealed to be incomplete when the next level's coursework introduces the replacement.

Consider the person who exits this pipeline at any point before its end. At age fifteen, they know the planetary atom and Newton's laws. At age twenty, they know the Bohr atom and some quantum formalism. At age twenty-two, they know quantum mechanics as a mathematical framework but may not know the Standard Model as a physical description of reality. At any exit point, this person has an incomplete, partially wrong understanding of what physics has actually established. Not because they are stupid. Not because they are lazy. Because the pipeline is designed to defer actual content for twenty years.

When this person — intelligent, curious, interested in fundamental questions — tries to engage with frontier physics, they work from the information they have. They propose theories that ignore quarks because they were never taught about quarks. They violate symmetries because they were never shown the symmetries. They contradict measurements because they were never given the measurements. They reinvent ideas that were tested and discarded decades ago because they were never told about the testing or the discarding.

The institution calls them cranks.

The institution built the pipeline that produced them.

This is not an abstract causal claim. The institution controls physics education. University physics departments set curriculum requirements. Physics professors write textbooks. Physics education researchers study pedagogy. National curriculum standards are influenced by the physics community. At every point in the pipeline, the institution had the option to teach the actual Standard Model — not the mathematics, which requires years of preparation, but the structural picture. What exists. What has been measured. What the measurements show.

The structural picture of the Standard Model is not inherently more difficult than the wrong models currently taught. A child who can learn “electrons orbit the nucleus” can learn “electrons are patterns of interaction that exist around nuclei through electromagnetic channels.” A child who can learn “gravity pulls things down” can learn “gravity is a drain toward the center of the parent body that applies at each moment.” A child who

can learn “there are protons and neutrons in the nucleus” can learn “protons and neutrons are patterns of three quarks bound by the strong force, and almost all of their mass comes from the binding energy, not from the quarks themselves.”

These descriptions are no more complex than the wrong descriptions currently taught. They are more accurate. They use vocabulary that connects to the actual Standard Model rather than vocabulary that must be unlearned before the Standard Model can be introduced. A child taught this vocabulary from age eight would arrive at age eighteen knowing what quarks are, what the four forces do, what the Higgs mechanism is, what neutrinos are and why they oscillate, what CP violation is, and what has been measured and to what precision. This person cannot be a crank in the traditional sense. They know too much about established results to ignore them. They might still be wrong about frontier questions. But they would be wrong productively — proposing mechanisms that are consistent with known measurements but differ from current models. That is not crankery. That is science.

The institution that complains about cranks manufactured the conditions that produce cranks. The institution that spends time and resources filtering out uninformed outsiders could instead spend a fraction of those resources ensuring that outsiders are informed. The educational pipeline is a choice. The choice to teach wrong models for twenty years before revealing the real ones is a choice. The consequence of that choice — a population of intelligent, curious people with catastrophically incomplete physics knowledge — is predictable and predicted. The crank problem is not an external affliction visited upon the institution by an ignorant public. It is the direct, predictable, manufactured consequence of the institution’s own educational choices.

7. The Lost Contributions

The double block — internal discouragement of commitment plus external exclusion of outsiders — has a cost. The cost cannot be precisely quantified because it consists of contributions that were never evaluated. The contributions that were never evaluated include some that would have been worthless, some that would have been wrong in productive ways, and some that would have advanced the field. The institution cannot distinguish these categories because it never examined the work. The cost is invisible by construction.

What can be described is the categories of contribution that the institutional structure specifically excludes, and why those categories are specifically needed.

The first excluded category is cross-domain structural methodology. Systems engineering, information theory, and software architecture train practitioners to ask questions that physics does not train for. “Is this problem generated by the domain or by the tools used to work in the domain?” is a standard systems engineering question. It is not a standard physics question. Yet it applies to physics directly. The class hierarchy problem in software — an artifact of programming languages that require class hierarchies, not an artifact of the data transformation being performed — has a precise analog in physics. The

“inherent complexity” of quantum field theory renormalization may be an artifact of continuous mathematics applied to a discrete system, not an artifact of the physics. A systems engineer who asks this question and pursues it methodically is applying a methodology the field needs. The crank label prevents the methodology from entering.

The second excluded category is engineering commitment culture. Engineers practice commitment daily. A structural engineer commits to specific load values, specific material properties, specific safety factors. The bridge either stands or falls. A software engineer commits to specific performance targets, specific correctness criteria, specific failure modes. The software either works or it doesn't. Being wrong in engineering is treated as a located failure to be investigated and corrected, not as a character flaw to be punished. This culture of commitment — where falsification is a normal daily event, where located errors are the primary mechanism of improvement, where being wrong at a specific point is more valuable than being vaguely right about nothing — is precisely what Popper described as the engine of scientific progress. Engineers practice it. Physicists describe it in philosophy seminars and do not practice it in their research. An outsider who brings engineering commitment culture to physics is bringing Popper's own epistemology in its practiced form. This is Popper's third and fourth elements — priority and acceptance — embodied in a professional culture rather than discussed in a philosophical text.

The third excluded category is computational substrate thinking. A software engineer who builds execution infrastructure — data transformation pipelines, bytecode interpreters, state machines, discrete update loops — thinks in terms of finite computation, discrete state, deterministic update, and exact arithmetic. When that engineer looks at physics and asks “what if the universe computes discretely?”, they are applying domain expertise that no physicist has. The physicist knows the formalism of continuous quantum field theory. The engineer knows what discrete computation actually looks like at the implementation level — what update loops do, how finite arithmetic behaves, why discrete systems don't produce the divergences that continuous formalisms produce. The question “why does the QED calculation diverge?” has a computational answer that a software engineer can see and a physicist trained exclusively in continuous mathematics cannot: the calculation diverges because continuous integration over unbounded domains diverges, and if the actual computation is discrete over bounded domains, the divergence is an artifact of the tool, not of the physics. This is a productive question. It leads to testable predictions. The crank label prevents it from being asked in a context where it could be evaluated.

The structural cost of the double block is not merely the loss of specific contributions. It is the loss of entire categories of thinking that the field needs and cannot generate internally because its training pipeline does not produce them. The institution trains physicists. Physicists think like physicists. The problems that remain unsolved in physics have resisted solution by people who think like physicists for decades. The possibility that these problems would yield to people who think differently — who bring structural methodology, commitment culture, and computational substrate expertise — is not considered because the people who think differently are excluded before their thinking can be examined.

8. The Falsification Conditions

This paper practices what it prescribes. Every major claim has an explicit failure condition. Each failure condition is observable. If the claim fails, the failure is this paper's contribution — a located error identifying where the structural analysis is wrong.

Claim 1: Status preservation has replaced falsification as the operating principle.

The test: examine the ratio of committed falsifiable predictions to hedged unfalsifiable claims in physics publications. If the ratio favors committed predictions — if the majority of published claims state specific values with explicit failure conditions — then this paper's characterization of the institution is wrong.

Claim 2: The refusal to commit matches the structural definition of cowardice.

The test: identify the available action (commitment to specific values), the collective benefit (located errors and accumulated platforms), and the personal risk (status loss from being publicly wrong). If commitment carries no personal risk in institutional physics — if being publicly wrong about a specific prediction has no career consequences — then the incentive mechanism this paper identifies does not exist and the cowardice characterization fails.

Claim 3: The crank label is applied without evaluating the work. The test: present work by an uncredentialed outsider that meets explicit quality criteria — falsifiable predictions, declared inputs, reproducible derivations, matches to established measurements — to institutional evaluation processes. If the work is evaluated on its content and accepted or rejected based on its content, then the credential-first filtering this paper describes does not operate as described.

Claim 4: The educational pipeline manufactures the crank problem. The test: implement Standard Model structural education from elementary school. If, after a generation, the rate of uninformed outsider engagement with frontier physics does not decrease, then the causal link between educational withholding and crank production does not exist.

Claim 5: If incentive reform, education reform, and evaluation reform all failed to increase the rate of parameter closure in physics — if none of these structural changes produced any measurable acceleration — then the thesis of this paper is wrong. The obstacles to closure are not status preservation, educational withholding, and credential gatekeeping. They are something else entirely, and this paper's structural analysis failed to identify them.

Each of these tests is specific. Each is observable. Each could falsify the corresponding claim. The paper commits to each claim and accepts falsification if it comes.

9. Closing

Falsification requires commitment. Commitment requires accepting the possibility of being wrong. Being wrong in a high-status field carries disproportionate career risk. The rational individual response to this risk is to avoid commitment. The aggregate of rational

individual responses is a field that has structurally ceased practicing falsification while continuing to claim it as an epistemological foundation.

This aggregate behavior pattern matches the structural definition of cowardice: an available action that would produce collective benefit is declined to avoid personal risk. The definition is mechanical. Its application is testable. The word is not an insult. It is a description of a decision pattern that is observable in the aggregate behavior of the institution.

The institution compounds internal non-commitment by blocking external contribution through an undifferentiated exclusion label that prevents evaluation of work on its merits. The label is applied based on credentials, not content. It violates Popper's criterion by evaluating the claimant rather than the claim. It intercepts work before examination, making its own accuracy untestable.

The institution further compounds the problem by withholding foundational physics education for twenty years, manufacturing a population of people with incomplete knowledge who then produce the uninformed engagement that the institution uses to justify the exclusion label. The institution builds the pipeline that produces the problem that justifies the gate that blocks the contributions that could advance the field that the institution claims to serve.

The solution is structural because the problem is structural. Change the incentive structure to reward commitment and located errors. Change the educational pipeline to teach the Standard Model from elementary school. Change the evaluation process to examine work on content before credentials. Each of these changes is implementable. Each has a predicted effect. Each predicted effect is testable.

The universe closes physics every Planck tick. It commits to specific values, updates deterministically, and produces exact results. It does not hedge. It does not average. It does not hold parameters open that observations have closed. It does not exclude inputs based on their institutional origin. It runs, and the result is what we measure.

Saying that physics cannot be closed is saying the universe cannot do what it does every tick. The institution's position that the domain is permanently open is not a falsifiable claim about physics. It is contextual data about the institution — a description of the institution's relationship to its own status, its own incentive structure, and its own educational choices. The universe does not consult the institution before updating. The universe runs.

References

- [1] K. Popper, *The Logic of Scientific Discovery*, 1934 (German), 1959 (English).
- [2] Particle Data Group, "Review of Particle Physics," *Progress of Theoretical and Experimental Physics*, 2022.
- [3] CODATA Task Group on Fundamental Constants, "CODATA Recommended Values of the Fundamental Physical Constants," *Reviews of Modern Physics*, 2021.

[4] E. Tiesinga et al., “CODATA 2018 values of the fundamental physical constants,” *Journal of Physical and Chemical Reference Data*, vol. 50, 2021.

Series cross-references (for in-depth treatment of concepts introduced in this paper):

- The distinction between falsifiable information and unfalsifiable contextual data: [HOWL-INFO-10-2026]
- Statistical breadth substituting for functional depth: [HOWL-CULT-1-2026]
- The institutional gap between departments: [HOWL-CULT-2-2026]
- The structural cessation of falsification practice: [HOWL-CULT-3-2026]
- What falsification structurally requires: [HOWL-CULT-4-2026]
- Why located errors are the most valuable finding: [HOWL-CULT-5-2026]
- Publications as immutable timestamps: [HOWL-CULT-6-2026]
- Inherited enforcement of untested norms: [HOWL-CULT-7-2026]
- Working in the space between departments: [HOWL-CULT-8-2026]
- Institutional structure for cross-domain work: [HOWL-CULT-9-2026]
- Departmental boundaries preventing recognition of unifying constants: [HOWL-CULT-10-2026]
- The taxonomy of engineering and science: [HOWL-CULT-11-2026]

[HOWL-CULT-12-2026] Cowardice and High Status as Anti-Popper Progress Blocking. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-12-2026>

[HOWL-CULT-1-2026] The N=1000 Fallacy. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-1-2026>

[HOWL-CULT-2-2026] The Domain Boundary. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-2-2026>

[HOWL-CULT-3-2026] The Falsification Forfeiture. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-3-2026>

[HOWL-CULT-4-2026] What Falsification Actually Requires. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-4-2026>

[HOWL-CULT-5-2026] Falsification as Finding. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-5-2026>

[[HOWL-CULT-6-2026](#)] Immutable and Mortal. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-6-2026>

[[HOWL-CULT-7-2026](#)] The Five Monkeys Problem. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-7-2026>

[[HOWL-CULT-8-2026](#)] The Omni-Domain Investigator. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-8-2026>

[[HOWL-CULT-9-2026](#)] The Omni-Academy. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-9-2026>

[[HOWL-INFO-10-2026](#)] Closing Domains. Github: <https://github.com/ghowland/papers/tree/main/papers/INFO/HOWL-INFO-10-2026>

[[HOWL-CULT-10-2026](#)] The Geometric Remainder Across Departments. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-10-2026>

[[HOWL-CULT-11-2026](#)] Engineering and Science. Github: <https://github.com/ghowland/papers/tree/main/papers/CULT/HOWL-CULT-11-2026>